

A Hierarchical Centered Dirichlet Process Mixture for Error Distributions in Bayesian Data Fusion under Unmeasured Confounding

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Abstract

We extend the Bayesian nonparametric framework for combining randomized and observational data under unmeasured confounding by allowing the residual distribution in the accelerated failure time outcome model to differ across data sources. Building on the centered Dirichlet process (CDP) construction of Yang et al. [2010], used in a single-source AFT-BART setting by Henderson et al. [2020], we place a hierarchical Dirichlet process (HDP) prior [Teh et al., 2006] on the source-specific mixing measures, with shared atoms but source-specific weights. The mean-zero constraint required for identifiability of the regression components is enforced separately within each source by recentering. We derive a blocked Gibbs sampler that preserves conjugacy with the BART updates of the prognostic, treatment-effect, and confounding-function components.

1 Motivation

In our framework for combining randomized and observational data under unmeasured confounding [Jacobs, 2026], the conditional expectation of the observed log-survival outcome admits the decomposition

$$\mathbb{E}[Y \mid A, X, S] = m_0(X, S) + \tau(X)A + (1 - S)Ac(X), \quad (1)$$

under SUTVA, randomization in the RCT, and transportability of potential outcomes. The corresponding individual-level model takes the form

$$Y_i = m_0(X_i, S_i) + A_i \tau(X_i) + A_i(1 - S_i)c(X_i) + W_i, \quad W_i \sim F_{S_i}, \quad (2)$$

where F_s is the error distribution within data source $s \in \{0, 1\}$ ($s = 1$ for the RCT, $s = 0$ for the observational study). For m_0 , τ , and c to be identified as the conditional-mean components defined in our framework, each F_s must have mean zero.

Henderson et al. [2020] model the residual distribution in a single-source AFT model as a location mixture of normals with a centered Dirichlet process prior on the mixing measure, leveraging the construction of Yang et al. [2010]. In their setting, a single CDP is sufficient because there is only one source. In ours, the natural question is whether F_0 and F_1 should be assumed equal. We argue they should not: real-world data and randomized trials typically differ in measurement quality, follow-up protocols, and the prevalence of extreme outcomes, and these differences plausibly enter the error distribution rather than the structural mean. At the same time, the two error distributions are unlikely to be entirely unrelated, and pooling information across them through a shared base measure is desirable.

A hierarchical Dirichlet process [Teh et al., 2006] resolves this tension: F_0 and F_1 share a common pool of latent “residual types” through a top-level random measure, while admitting source-specific weights. Combined with the centering construction of Yang et al. [2010], applied within each source, this yields source-specific, mean-zero, nonparametric error distributions that borrow information through their shared atoms.

2 Notation and observed data

We observe data $\mathcal{D} = \{(Y_i^{\text{obs}}, \delta_i, A_i, X_i, S_i) : i = 1, \dots, n\}$, where $Y_i^{\text{obs}} = \min(Y_i, \log C_i)$ is the log of the observed time, $\delta_i = \mathbf{1}\{Y_i \leq \log C_i\}$ is the event indicator, $A_i \in \{0, 1\}$ is treatment, $X_i \in \mathbb{R}^p$ is a vector of observed covariates, and $S_i \in \{0, 1\}$ is the source indicator. Write $n_s = \sum_i \mathbf{1}\{S_i = s\}$.

The model is (2). The latent complete-data log-survival time Y_i equals Y_i^{obs} if $\delta_i = 1$ and is unobserved if $\delta_i = 0$, in which case we know $Y_i > \log C_i$.

3 Error model

3.1 Per-source location mixture of Gaussians

Conditional on the source $S_i = s$, we model the residual as a location mixture of Gaussians with common scale $\sigma > 0$ and source-specific mixing measure G_s :

$$W_i | S_i = s, G_s, \sigma \sim f_s(w) = \int \frac{1}{\sigma} \phi\left(\frac{w - \theta}{\sigma}\right) dG_s(\theta). \quad (3)$$

Introducing a latent location θ_i for each observation gives the hierarchical representation

$$W_i | \theta_i, \sigma^2 \sim N(\theta_i, \sigma^2), \quad \theta_i | G_{S_i} \sim G_{S_i}. \quad (4)$$

The mean-zero constraint $\mathbb{E}[W_i | S_i = s, G_s, \sigma] = 0$ is enforced by requiring $\int \theta dG_s(\theta) = 0$ for both s .

A shared scale across sources is the most parsimonious choice and parallels Henderson et al. [2020]; a source-specific σ_s requires only a one-line change and is discussed in Section 7.

3.2 Hierarchical Dirichlet process prior on the mixing measures

We place a hierarchical Dirichlet process prior [Teh et al., 2006] on (G_0^*, G_1^*) via a top-level random measure G^* :

$$G^* \sim \text{DP}(\gamma, H), \quad (5)$$

$$G_s^* | G^*, M_s \sim \text{DP}(M_s, G^*), \quad s \in \{0, 1\}, \quad (6)$$

where $H = N(0, \sigma_\theta^2)$ is the base distribution on $\mathbb{R}$, $\gamma > 0$ is the top-level concentration, and $M_0, M_1 > 0$ are source-specific concentrations. We refer to G_s^* as the *unconstrained* version of G_s ; the centering described in Section 3.3 produces G_s .

By Sethuraman’s stick-breaking representation [Sethuraman, 1994], the top-level measure can be written as

$$G^* = \sum_{k=1}^{\infty} \beta_k \delta_{\theta_k^*}, \quad \theta_k^* \stackrel{\text{iid}}{\sim} H, \quad \beta_k = u_k \prod_{l < k} (1 - u_l), \quad u_k \stackrel{\text{iid}}{\sim} \text{Beta}(1, \gamma). \quad (7)$$

Conditional on G^* , each G_s^* shares the same atoms but has its own weights:

$$G_s^* = \sum_{k=1}^{\infty} \pi_{sk} \delta_{\theta_k^*}, \quad (8)$$

with the source-specific weight vector $\boldsymbol{\pi}_s = (\pi_{sk})_{k \geq 1}$ distributed according to the conditional law $\text{DP}(M_s, G^*)$ projected onto the atoms.

3.3 Mean-zero centering

For each source s , define the mean of the unconstrained measure:

$$\mu_s = \int \theta dG_s^*(\theta) = \sum_{k=1}^{\infty} \pi_{sk} \theta_k^*. \quad (9)$$

Following Yang et al. [2010], the *centered* measure is

$$G_s = \sum_{k=1}^{\infty} \pi_{sk} \delta_{\theta_k^* - \mu_s}. \quad (10)$$

By construction $\int \theta dG_s(\theta) = 0$ almost surely, so (3) satisfies $\mathbb{E}[W_i | S_i = s, G_s, \sigma] = 0$ as required.

Although the unconstrained atoms θ_k^* are shared between sources, the centered atoms $\theta_{sk} := \theta_k^* - \mu_s$ are source-specific: they share their unconstrained location θ_k^* but are shifted by source-specific amounts μ_0 and μ_1 . This is the precise sense in which the two error distributions borrow information while retaining flexibility: the *shapes* of F_0 and F_1 are linked through the shared θ_k^* , while the *weights* on those shapes (and the resulting mean shifts) are source-specific.

3.4 Truncation for posterior computation

For computation we truncate the top-level stick-breaking at a finite level K [Ishwaran and James, 2001]: $u_K = 1$, so $\beta_K = 1 - \sum_{k < K} \beta_k$ and $\beta_k = 0$ for $k > K$. Conditional on the truncated $\boldsymbol{\beta} \in \Delta^{K-1}$ we use the finite-dimensional approximation

$$\boldsymbol{\pi}_s | \boldsymbol{\beta}, M_s \sim \text{Dirichlet}(M_s \beta_1, \dots, M_s \beta_K), \quad (11)$$

which is the standard truncated representation of the HDP [Teh et al., 2006, Wang et al., 2009]; the approximation becomes exact as $K \rightarrow \infty$. We monitor the largest occupied component index across iterations and increase K if it approaches the truncation bound (default $K = 50$, following Henderson et al. [2020]).

4 Full hierarchical model

Introducing latent cluster assignments $Z_i \in \{1, \dots, K\}$ such that $\theta_i = \theta_{Z_i}^* - \mu_{S_i}$, the full hierarchical model is:

$$Y_i | Z_i, \boldsymbol{\theta}^*, \sigma^2, m_0, \tau, c \sim N(m_0(X_i, S_i) + A_i \tau(X_i) + A_i(1 - S_i) c(X_i) + \theta_{Z_i}^* - \mu_{S_i}, \sigma^2), \quad (12)$$

$$Z_i | \boldsymbol{\pi}_{S_i} \sim \text{Categorical}(\boldsymbol{\pi}_{S_i}), \quad (13)$$

$$\boldsymbol{\pi}_s | \boldsymbol{\beta}, M_s \sim \text{Dirichlet}(M_s \boldsymbol{\beta}_1, \dots, M_s \boldsymbol{\beta}_K), \quad s \in \{0, 1\}, \quad (14)$$

$$\boldsymbol{\beta} | \gamma \sim \text{Stick}_K(\gamma), \quad (15)$$

$$\theta_k^* \stackrel{\text{iid}}{\sim} N(0, \sigma_\theta^2), \quad k = 1, \dots, K, \quad (16)$$

$$m_0, \tau, c \sim \text{BART}, \quad (17)$$

$$\sigma^2 \sim \kappa \nu / \chi_\nu^2, \quad (18)$$

$$\gamma \sim \text{Gamma}(a_\gamma, b_\gamma), \quad (19)$$

$$M_s \sim \text{Gamma}(a_M, b_M), \quad s \in \{0, 1\}, \quad (20)$$

with $\mu_s = \sum_k \pi_{sk} \theta_k^*$ a deterministic function of $\boldsymbol{\pi}_s$ and $\boldsymbol{\theta}^*$. Here $\text{Stick}_K(\gamma)$ denotes the truncated stick-breaking distribution with $\text{Beta}(1, \gamma)$ breaks.

For the BART components we follow Chipman et al. [2010] with the AFT-specific adaptations of Henderson et al. [2020]: $J = 200$ trees, $\alpha = 0.95$, $\beta = 2$, response transformation $y_i^{\text{tr}} = y_i \exp(-\hat{\mu}_{\text{AFT}})$ based on a preliminary parametric log-normal AFT fit, and calibration of κ , σ_θ^2 via the Yamato-type approximation described in Henderson et al. [2020, Section 2.4]. For the concentration hyperpriors we set $a_\gamma = a_M = 2$, $b_\gamma = b_M = 0.1$, mirroring Henderson’s defaults; the resulting prior on each concentration has mean 20 and prior mode 10.

5 Posterior computation via blocked Gibbs sampling

Let $\boldsymbol{\Theta}$ denote the full state of the sampler:

- BART parameters (trees and node values) for m_0, τ, c ;
- Cluster assignments $\mathbf{Z} = (Z_1, \dots, Z_n)$;
- Unconstrained atoms $\boldsymbol{\theta}^* = (\theta_1^*, \dots, \theta_K^*)$;
- Top-level stick parameters $\mathbf{u}$ (equivalently $\boldsymbol{\beta}$);
- Source-specific weights $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1$;
- Concentration parameters γ, M_0, M_1 ;
- Error scale σ^2 ;
- Imputed log-survival times $\{Y_i : \delta_i = 0\}$.

The source means $\boldsymbol{\mu} = (\mu_0, \mu_1)$ are recomputed deterministically as $\mu_s = \sum_k \pi_{sk} \theta_k^*$ whenever $\boldsymbol{\pi}$ or $\boldsymbol{\theta}^*$ change.

For notational economy, define the *partial residual* for individual i :

$$R_i := Y_i - m_0(X_i, S_i) - A_i \tau(X_i) - A_i(1 - S_i) c(X_i). \quad (21)$$

Conditional on the BART components and $Z_i = k$, we have $R_i \sim N(\theta_k^* - \mu_{S_i}, \sigma^2)$.

One sweep of the Gibbs sampler updates each block in turn. We describe the steps in the order in which they should be executed.

5.1 Step 1: Data augmentation for censored observations

For each i with $\delta_i = 0$, draw the imputed log-survival time

$$Y_i | \cdot \sim \text{TruncatedNormal}(\eta_i, \sigma^2; [\log C_i, \infty)), \quad (22)$$

where $\eta_i = m_0(X_i, S_i) + A_i \tau(X_i) + A_i(1 - S_i) c(X_i) + \theta_{Z_i}^* - \mu_{S_i}$ is the current conditional mean of Y_i .

5.2 Step 2: BART backfitting

The three BART components are updated sequentially via Bayesian backfitting [Chipman et al., 2010]. For each component, the working response is the partial residual obtained by subtracting the contributions of all other components from Y_i , along with the current cluster shift $\theta_{Z_i}^* - \mu_{S_i}$.

Update m_0 . Working response:

$$R_i^{(m_0)} = Y_i - A_i \tau(X_i) - A_i(1 - S_i) c(X_i) - (\theta_{Z_i}^* - \mu_{S_i}). \quad (23)$$

Treat $R_i^{(m_0)} \sim N(m_0(X_i, S_i), \sigma^2)$ and apply the standard MH tree moves and conjugate node-value updates on the predictor vector (X_i, S_i) .

Update τ . Active only for treated observations ($A_i = 1$). Working response:

$$R_i^{(\tau)} = Y_i - m_0(X_i, S_i) - (1 - S_i) c(X_i) - (\theta_{Z_i}^* - \mu_{S_i}), \quad i \text{ with } A_i = 1. \quad (24)$$

Treat $R_i^{(\tau)} \sim N(\tau(X_i), \sigma^2)$ and update the BART for τ on this subset. Control observations carry no information about τ .

Update c . Active only for treated observations in the OS ($A_i = 1, S_i = 0$). Working response:

$$R_i^{(c)} = Y_i - m_0(X_i, 0) - \tau(X_i) - (\theta_{Z_i}^* - \mu_0), \quad i \text{ with } A_i = 1, S_i = 0. \quad (25)$$

Treat $R_i^{(c)} \sim N(c(X_i), \sigma^2)$ and update the BART for c on this subset.

After updating all three BART components, recompute R_i as in (21) for use in the remaining steps.

5.3 Step 3: Cluster assignments

For each i , sample $Z_i \in \{1, \dots, K\}$ from

$$P(Z_i = k | \cdot) \propto \pi_{S_i, k} \phi\left(\frac{R_i - (\theta_k^* - \mu_{S_i})}{\sigma}\right), \quad k = 1, \dots, K. \quad (26)$$

5.4 Step 4: Unconstrained atoms

Following Henderson et al. [2020], atoms are updated in the unconstrained parameterization and the centering is reapplied deterministically afterwards. Let $n_k = \sum_i \mathbf{1}\{Z_i = k\}$ be the total number of observations in cluster k across both sources, and define the cluster sum $\bar{R}_k = \sum_{i:Z_i=k} R_i$.

The conjugate Gaussian update is

$$\theta_k^* | \cdot \sim N\left(\frac{\sigma^{-2} \bar{R}_k}{\sigma_\theta^{-2} + n_k \sigma^{-2}}, \frac{1}{\sigma_\theta^{-2} + n_k \sigma^{-2}}\right), \quad k = 1, \dots, K. \quad (27)$$

After updating all θ_k^* , recompute the source means and centered atoms:

$$\mu_s = \sum_{k=1}^K \pi_{sk} \theta_k^*, \quad \theta_{sk} = \theta_k^* - \mu_s, \quad s \in \{0, 1\}. \quad (28)$$

Remark. The update (27) is the standard Gibbs step for the unconstrained mixing measure; the centering acts as a deterministic transformation between an unidentified parameterization $(m_0^{\text{unc}}, \theta^*)$ and the identified (m_0, θ) . As discussed by Yang et al. [2010] and Henderson et al. [2020], this parameter-expansion strategy preserves the correct marginal posterior for the centered quantities.

5.5 Step 5: Top-level stick weights

The HDP update for the top-level stick-breaking weights uses the auxiliary table-count augmentation of Teh et al. [2006], originally derived from Antoniak [1974].

Step 5a: Sample table counts. Let $n_{sk} = \sum_i \mathbf{1}\{S_i = s, Z_i = k\}$. For each (s, k) with $n_{sk} > 0$, sample the auxiliary table count $t_{sk} \in \{1, \dots, n_{sk}\}$. The marginal distribution is

$$P(t_{sk} = t | \cdot) \propto |s(n_{sk}, t)| (M_s \beta_k)^t, \quad (29)$$

where $|s(n, t)|$ are unsigned Stirling numbers of the first kind. In practice t_{sk} is generated via the equivalent representation

$$t_{sk} = \sum_{j=1}^{n_{sk}} B_{sk,j}, \quad B_{sk,j} \sim \text{Bernoulli}\left(\frac{M_s \beta_k}{M_s \beta_k + j - 1}\right), \quad (30)$$

which avoids the Stirling-number evaluation. Set $t_{sk} = 0$ if $n_{sk} = 0$.

Step 5b: Sample stick weights. Let $t_{\cdot k} = t_{0k} + t_{1k}$. Sample the truncated sticks

$$u_k | \cdot \sim \text{Beta}\left(1 + t_{\cdot k}, \gamma + \sum_{l>k} t_{\cdot l}\right), \quad k = 1, \dots, K-1, \quad (31)$$

and set $u_K = 1$. Recompute $\beta_k = u_k \prod_{l<k} (1 - u_l)$ for $k = 1, \dots, K$.

5.6 Step 6: Source-specific weights

Given β and the cluster assignments, π_s has a conjugate Dirichlet update:

$$\pi_s | \cdot \sim \text{Dirichlet}(M_s \beta_1 + n_{s1}, \dots, M_s \beta_K + n_{sK}), \quad s \in \{0, 1\}. \quad (32)$$

After this step, recompute μ_s and $\theta_{sk} = \theta_k^* - \mu_s$.

5.7 Step 7: Concentration parameters

Both γ and M_s are updated using the auxiliary-variable approach of Escobar and West [1995].

Update γ . Let K^* denote the number of occupied top-level components and let $t_{..} = \sum_{s,k} t_{sk}$ denote the total number of tables across both restaurants.

1. Sample $\eta_\gamma \mid \gamma, t_{..} \sim \text{Beta}(\gamma + 1, t_{..})$.
2. Sample $\gamma \mid \eta_\gamma, K^*$ from the two-component mixture

$$\begin{aligned} \gamma \mid \cdot &\sim \omega_\gamma \text{Gamma}(a_\gamma + K^*, b_\gamma - \log \eta_\gamma) \\ &\quad + (1 - \omega_\gamma) \text{Gamma}(a_\gamma + K^* - 1, b_\gamma - \log \eta_\gamma), \end{aligned}$$

with

$$\frac{\omega_\gamma}{1 - \omega_\gamma} = \frac{a_\gamma + K^* - 1}{t_{..}(b_\gamma - \log \eta_\gamma)}.$$

Update M_s . For each $s \in \{0, 1\}$, with $t_s = \sum_k t_{sk}$ source-specific table counts:

1. Sample $\eta_{M,s} \mid M_s, n_s \sim \text{Beta}(M_s + 1, n_s)$.
2. Sample $M_s \mid \eta_{M,s}, t_s$ from the analogous two-component Gamma mixture with t_s in place of K^* and n_s implicit in $\eta_{M,s}$.

See Teh et al. [2006, Appendix A] for the full derivation in the HDP setting.

5.8 Step 8: Error scale

Conditional on assignments, atoms, and BART components,

$$\sigma^2 \mid \cdot \sim \text{InverseGamma}\left(\frac{\nu + n}{2}, \frac{\kappa\nu + \hat{s}^2}{2}\right), \quad \hat{s}^2 = \sum_{i=1}^n (R_i - (\theta_{Z_i}^* - \mu_{S_i}))^2. \quad (33)$$

5.9 Algorithmic summary

One sweep of the Gibbs sampler consists of:

- (1) Impute Y_i for censored i (Step 1).
- (2) Update the BART components m_0 , then τ , then c , recomputing R_i at the end (Step 2).
- (3) Update cluster assignments $\mathbf{Z}$ (Step 3).
- (4) Update the unconstrained atoms $\boldsymbol{\theta}^*$ and recompute μ_s, θ_{sk} (Step 4).
- (5) Update top-level table counts $\{t_{sk}\}$ and sticks $\mathbf{u}$, then $\boldsymbol{\beta}$ (Step 5).
- (6) Update source weights $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1$ and recompute μ_s, θ_{sk} (Step 6).
- (7) Update concentration parameters γ, M_0, M_1 (Step 7).
- (8) Update σ^2 (Step 8).

6 Practical considerations

Truncation. The default $K = 50$ follows Henderson et al. [2020]. Monitor $\max_i Z_i$ across iterations; if it equals K in more than a small fraction of iterations, K is too small.

Hyperparameter calibration. The induced prior on $\text{Var}(W | S = s)$ under the HDP-CDP has the same structural form as in the single-source case of Henderson et al. [2020], with source-specific contributions. The Yamato-type approximation [Yamato, 1984] they use to obtain a tractable approximation for $\sum_k \pi_{sk} (\theta_k^* - \mu_s)^2 / \sigma_\theta^2$ extends without modification to each source separately. Calibration to a preliminary $\hat{\sigma}_W^2$ obtained from a parametric log-normal AFT fit is recommended, with default tail probability $q = 0.5$.

Initialization. Initialize the BART components via the standard Chipman et al. [2010] initialization; assign cluster labels by k -means on residuals from an initial parametric fit; set $\beta = (1/K, \dots, 1/K)$, $\pi_s = \beta$, $\gamma = M_0 = M_1 = 1$.

Diagnostics. The source-specific means μ_0, μ_1 and the variances $\text{Var}(W | S = s)$ are derived quantities of the posterior and serve as natural diagnostics of how much source-specific structure the HDP is exploiting. Posterior summaries of $|\mu_0 - \mu_1|$ and of the source-specific residual variances communicate the extent to which F_0 and F_1 differ in their first two moments; the full residual densities $\hat{f}_s(w) = K^{-1} \sum_k \pi_{sk} \phi((w - \theta_{sk})/\sigma)/\sigma$ can be plotted for direct visual comparison.

7 Extensions

Several extensions follow naturally from this construction.

Source-specific scales. Replacing σ by σ_s requires only stratifying Steps 3 and 8 by source. This is appropriate when RWD has notably different noise magnitude than the RCT.

Nested rather than hierarchical Dirichlet process. A nested Dirichlet process [Rodríguez et al., 2008] on (G_0^*, G_1^*) would cluster the residual *distributions themselves*, identifying subpopulations within the OS whose residual structure matches the RCT and subpopulations whose does not. This is thematically aligned with the confounding-function decomposition: the bias function $c(X)$ absorbs the conditional-mean shift, and the nested DP would absorb the residual-shape shift.

Covariate-dependent error distribution. A dependent Dirichlet process [MacEachern, 1999] with covariate-dependent weights would extend the construction to fully heteroscedastic errors. Combining with the heteroskedastic AFT-BART approach of Sparapani et al. [2023], in which a second BART models $\log \sigma(X)$, yields an even more flexible alternative.

Connection to Bayesian borrowing diagnostics. Source-specific concentration parameters M_s , jointly with the source-specific table counts t_s , produced as a by-product of the sampler, function as effective-sample-size-style summaries of how much the OS-specific residual distribution borrows from the shared base measure G^* . This connects the construction to the broader Bayesian-borrowing literature on dynamic information-sharing diagnostics [Hupf et al., 2021].

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